

Topic 1.5 - Polynomial Functions and Complex Zeros

Practice Worksheet

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. In context, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Apply the Fundamental Theorem of Algebra and conjugate-pair theorem to count zeros.
- Construct real-coefficient polynomials from real and complex zeros.
- Connect real-zero multiplicity to graph behavior while distinguishing complex zeros from intercepts.

Section A: Foundation and representation

Question 1

Counting

No calculator

A real degree-6 polynomial has zeros $2 + i$, $-3i$, and 5. List required conjugates and the number of remaining zeros.

Question 2

Multiplicity

No calculator

A real degree-8 polynomial has zeros 1 multiplicity 3, $-2 + 4i$, and $5i$. How many zeros remain after conjugates?

Section A continued

Question 3

Construction

No calculator

Construct the monic cubic with zeros -4 and $2 \pm i$.

Question 4

Complex factoring

No calculator

Factor $x^4 + 5x^2 + 4$ completely over the complex numbers.

Question 5

Graph behavior

No calculator

Describe the real graph behavior at every real zero of $(x + 2)^3(x - 1)^2(x^2 + 9)$.

Section B: Application and reasoning

Question 6

Parameter

No calculator

If $P(x) = x^3 + kx^2 + x - 2$ has real coefficients and i is a zero, determine k and all zeros.

Question 7

Reasoning

No calculator

Show that $x^2 - 4x + 13$ has no real zeros and state its complex zeros.

Question 8

Degree

No calculator

Find the least possible degree of a real-coefficient polynomial having zeros 2, $1 + i$, and $-3i$.

Section C: Challenge and error analysis

Question 9

Error analysis

No calculator

A student marks $2 + i$ as an x -intercept. Diagnose the error and explain how that zero influences a real polynomial.

Question 10

Challenge

No calculator

Construct a quartic with real coefficients, zeros $1 \pm i$, $2 \pm 3i$, and leading coefficient -1 .

Question 11

Vieta

No calculator

For $P = x^3 + x^2 - 7x - 15$, use Vieta's relationships to state the sum and product of all zeros and verify with $3, -2 \pm i$.

Question 12

Challenge

No calculator

Give a degree-5 real polynomial with exactly one distinct real zero and four nonreal zeros. State multiplicities.

Section D: AP-style multiple choice

MCQ 1

No calculator

A real polynomial has zero $3 - 2i$. Which must also be a zero?

- A. $-3 + 2i$
- B. $3 + 2i$
- C. $-3 - 2i$
- D. $2 + 3i$

MCQ 2

No calculator

Which quadratic has zeros $-1 \pm 4i$?

- A. $x^2 - 2x + 17$
- B. $x^2 + 2x + 17$
- C. $x^2 + 2x - 15$
- D. $x^2 - 2x - 15$

MCQ 3

No calculator

A degree-7 real polynomial has three nonreal conjugate pairs. What must be true?

- A. It has no real zeros.
- B. It has at least one real zero.
- C. All real zeros have even multiplicity.
- D. It has exactly seven intercepts.

MCQ 4

No calculator

At a real zero of multiplicity 4, a polynomial graph generally

- A. crosses and changes sign
- B. touches and keeps sign
- C. has a vertical asymptote
- D. has a hole

AP-style free response 1**No calculator - 6 points**

A monic degree-5 polynomial with real coefficients has zeros -2 , $1 + 2i$, and 3 with multiplicity 2.

A. List all zeros with multiplicity.

B. Write factored form over the reals.

C. Expand the polynomial.

D. Describe graph behavior at its real zeros and end behavior.

AP-style free response 2**Calculator permitted - 6 points**

Consider $Q(x) = x^5 - 6x^4 + 11x^3 + 16x^2 - 50x + 52$.

A. Use technology to approximate all zeros.

B. Group the zeros into real zeros and conjugate pairs.

C. Write an exact factored form over the reals suggested by the approximations.

D. Explain why the graph has only one x -intercept.
